Dose Calibrator

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1 Objective

To perform a Quality Assurance (QA) test of the dose calibrator.

2 Apparatus

Equipment required:

- Dose calibrator
- Check source
- Canister
- Source holder

3 Theory

The dose calibrator is one of the most essential instruments in nuclear medicine for measuring and verifying the radioactivity of radiopharmaceuticals prior to patient administration [1]. A typical radioisotope calibrator comprises an ionization chamber, a high-voltage power supply, a sensitive electronic electrometer, and a display unit that allows the selection of the specific radioisotope calibration factor.

The ionization chamber utilized in dose calibrators is generally a well-type cylindrical design, which frequently utilizes Argon gas under high pressure ($\sim$ 20–30 atm) [2]. The geometrical configuration of the well-type design is engineered to closely approximate a 4π detection geometry, allowing it to capture the majority of the photon flux emitted by a source. The elevated gas pressure dramatically increases the density of the interactive medium, enhancing the probability of photon interaction and thereby providing superior detection efficiency. The calibrator relies on two coaxial cylindrical electrodes that are maintained under a steady accelerating electric field from a built-in power supply.

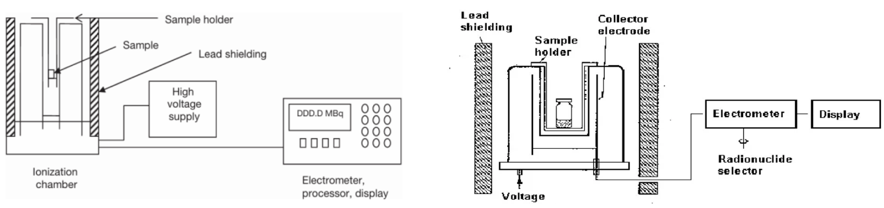

Figure 1: Schematic diagram of a typical dose calibrator. The radioactive source is placed in the well of the ionization chamber, where emitted photons interact with the gas to produce ion pairs. The resulting current is processed by the electrometer and displayed as activity in mCi or MBq. The lead shielding minimizes background radiation and protects personnel.

When a radioactive source (vial or syringe) is positioned inside the well, the emitted photons interact with the Argon gas chamber walls and the gas itself, primarily through the photoelectric effect and Compton scattering [3]. The resulting secondary electrons ionize the surrounding Argon atoms, creating ion pairs. Because the applied voltage places the chamber in the "ionization region" of the gas-filled detector characteristic curve, recombination of ion pairs is minimized, but no gas multiplication (such as Townsend avalanche) occurs [2]. The ions migrate toward the respective electrodes (anode and cathode), collecting a steady state current proportional to the total radioactivity. Since this current is exceptionally small (typically in the pA to μA range), a highly sensitive and stable electrometer is required to process and digitize the signal. The processed output is scaled by an isotope-specific calibration factor and displayed in functional units like milliCuries (mCi) or MegaBecquerels (MBq).

Surrounding the ionization chamber, an effective lead shielding is maintained, which not only mitigates personnel radiation exposure hazards but fundamentally suppresses background scatter and environmental radiation contributions [1]. A sample holder and a removable liner are routinely fitted, significantly facilitating ease of operation and decontamination in case of radiopharmaceutical spills.

3.1 Quality Assurance Tests

Strict adherence to quality assurance (QA) protocols is critical for the reliable and safe operation of a dose calibrator in a clinical setting. International regulatory associations such as the IAEA and AAPM have well-established recommendations for the minimum functional testing regimen [4, 5].

3.1.1 Accuracy Test

The accuracy test validates the degree of closeness of the measured activity to an accepted standard value. A calibrated standard reference source, prominently providing known activity, isotope type, and a reference time, is utilized. Typical test sources include long-lived isotopes like ^{137}Cs or ^{57}Co. The decay-corrected true activity at the time of the test is computed using the radioactive decay formula and then compared to the electrometer's measurement. As per international protocols, measured values must strictly fall within $\pm5\%$ of the theoretical standard value [4, 5].

3.1.2 Constancy Test

Constancy is a routine (typically daily) quality control check evaluating the stability and reproducibility of the dose calibrator against systematic drift. A long-lived source such as ^{137}Cs is repeatedly measured across the standard functional operating conditions. The fluctuation between repeated measurements should not diverge significantly, strictly remaining within a $\pm5\%$ allowable tolerance relative to reference baseline levels [4].

3.1.3 Linearity Test

The linearity test evaluates whether the instrument can accurately and linearly measure activities sequentially ranging over the vast dynamic scales encountered in medical environments (e.g., from maximal bolus activities in mCi down to negligible activities in μCi). Typically, a highly active short-lived source like ^{99m}Tc or ^{18}F-FDG is assessed iteratively over significant multiples of its half-life [1]. The empirically evaluated activities are tested against the expected mathematical decay function, establishing that the detector electrometer response introduces minimal deviation irrespective of saturation or low-scale noise limitations.

3.1.4 Geometry Test

The geometry test assesses the potential influence of spatial distribution and physical characteristics of the contained liquid volume on the overall recorded activity reading. Because of differential path lengths and self-absorption, readings theoretically can fluctuate depending on whether the source is densely packed in a small volume or scattered across a larger volume. The procedure essentially tests a uniform activity pool consecutively diluted with saline in identical syringe or vial formats [5]. For a properly calibrated chamber approximating 4π geometry homogeneously, systematic fluctuation recorded in activity due to geometrical configuration must consistently remain within an acceptable margin (normally under $\pm5\%$).

4 Observation

4.1 Tabulation for accuracy test (Source: ^{137}Cs gel source):

Initial Activity	Decay Corrected Activity	BKG	Measured Activity	Net Activity	% Deviation	Tolerance
0.211 mCi (18/02/2018)	174.1084591 μCi	1.8 μCi	172.5 μCi	170.7 μCi	1.957664252%	5%

4.2 Tabulation for constancy test (Source: ^{137}Cs):

Initial activity measured = 172.5 μCi

Figure 2: Experimental setup for geometry test. The source vial is placed in the well of the dose calibrator, and saline is added incrementally to assess the influence of geometry on the measured activity. The lead shielding around the chamber minimizes background radiation, ensuring accurate readings.

Repetition No.	Activity Measured (μCi)	Deviation (%)	Tolerance
1	172.5	0.000	2%
2	172.2	0.174	2%
3	172.5	0.000	2%
4	172.4	0.058	2%
5	171.9	0.349	2%
6	172.6	0.058	2%
7	171.6	0.524	2%
8	172.1	0.232	2%
9	172.4	0.058	2%
10	172.5	0.000	2%

4.3 Tabulation for linearity test (Source: ^{18}F-FDG, interval = 30 min):

Table 1: Auto Linearity Test (F-18, Interval = 30 min)

Date & Time	Elapsed (min)	Measured	Predicted	% Var
Apr 20 2026 15:42	0	9.29 mCi	9.29 mCi	0.0
Apr 20 2026 16:12	30	7.69 mCi	7.69 mCi	-0.0
Apr 20 2026 16:42	60	6.36 mCi	6.36 mCi	0.0
Apr 20 2026 17:12	90	5.26 mCi	5.26 mCi	-0.0
Apr 20 2026 17:42	120	4.36 mCi	4.36 mCi	0.0
Apr 20 2026 18:12	150	3.60 mCi	3.60 mCi	-0.1
Apr 20 2026 18:42	180	2.98 mCi	2.98 mCi	0.1
Apr 20 2026 19:12	210	2.47 mCi	2.47 mCi	-0.0
Apr 20 2026 19:42	240	2.04 mCi	2.04 mCi	0.1
Apr 20 2026 20:12	270	1.690 mCi	1.689 mCi	0.1
Apr 20 2026 20:42	300	1.396 mCi	1.397 mCi	-0.1
Apr 20 2026 21:12	330	1.155 mCi	1.156 mCi	-0.1
Apr 20 2026 21:42	360	956 uCi	956 uCi	-0.1
Apr 20 2026 22:12	390	791 uCi	791 uCi	-0.0
Apr 20 2026 22:42	420	654 uCi	655 uCi	-0.1
Apr 20 2026 23:12	450	541 uCi	542 uCi	-0.1
Apr 20 2026 23:42	480	448 uCi	448 uCi	-0.1
Apr 21 2026 00:12	510	370 uCi	371 uCi	-0.1
Apr 21 2026 00:42	540	307 uCi	307 uCi	-0.1

Auto Linearity Test (F-18, Interval = 30 min) (Continued)

Date & Time	Elapsed (min)	Measured	Predicted	% Var
Apr 21 2026 01:12	570	253 uCi	254 uCi	-0.2
Apr 21 2026 01:42	600	210 uCi	210 uCi	0.1
Apr 21 2026 02:12	630	173.8 uCi	173.8 uCi	0.0
Apr 21 2026 02:42	660	143.8 uCi	143.8 uCi	0.1
Apr 21 2026 03:12	690	118.8 uCi	118.9 uCi	-0.1
Apr 21 2026 03:42	720	98.4 uCi	98.4 uCi	0.0
Apr 21 2026 04:12	750	81.5 uCi	81.4 uCi	0.1
Apr 21 2026 04:42	780	67.6 uCi	67.4 uCi	0.4
Apr 21 2026 05:12	810	56.0 uCi	55.7 uCi	0.4
Apr 21 2026 05:42	840	46.5 uCi	46.1 uCi	0.7
Apr 21 2026 06:12	870	38.4 uCi	38.2 uCi	0.6
Apr 21 2026 06:42	900	31.8 uCi	31.6 uCi	0.7
Apr 21 2026 07:12	930	26.4 uCi	26.1 uCi	1.1
Apr 21 2026 07:42	960	21.9 uCi	21.6 uCi	1.2
Apr 21 2026 08:12	990	18.04 uCi	17.88 uCi	0.9
Apr 21 2026 08:42	1020	15.09 uCi	14.79 uCi	2.0

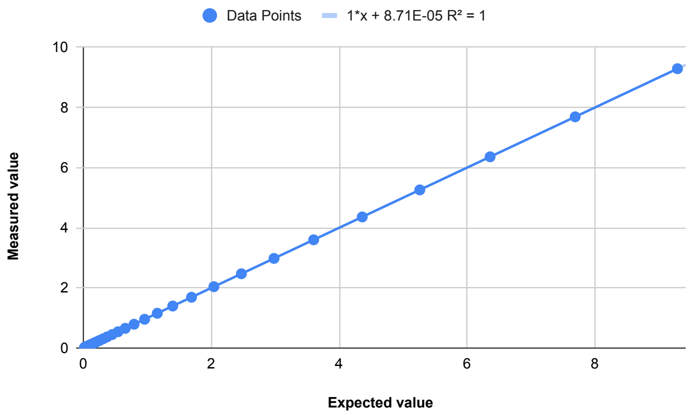

Figure 3: Linearity test plot for ^{18}F-FDG source. The measured activity closely follows the predicted decay curve, indicating good linearity of the dose calibrator across a wide range of activities.

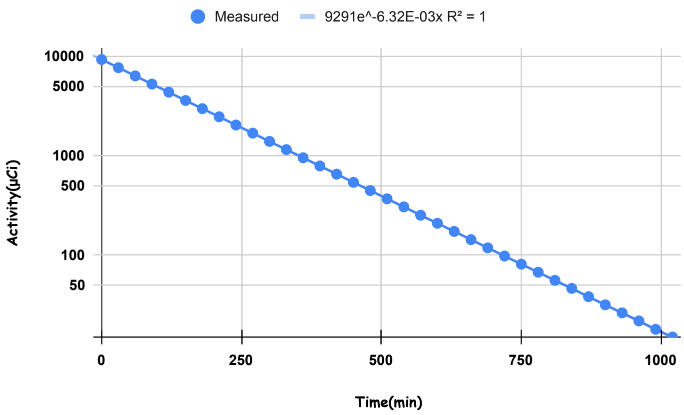

Figure 4: The time-activity curve for decay method.

4.4 Tabulation for geometry test (Source: ^{18}F):

Source F-18
Activity with only F-18 in Syringe = 1.29 mCi
Activity without F-18 in Syringe = 86 μCi
Background Activity = 23.5 μCi

Net Activity in Vial with F-18 only = 1.18 mCi

Saline Volume (ml)	Activity (mCi)	% Variation
1	1.19	+0.85
2	1.18	0.00
3	1.19	+0.85
4	1.17	-0.85
5	1.18	0.00
6	1.15	-2.54
7	1.16	-1.69
8	1.15	-2.54
9	1.14	-3.39
10	1.15	-2.54

Table 2: Percentage variation of activity with respect to 1.18 mCi

4.5 Result

The dose calibrator readings were within tolerance in the performed QA checks:

- Accuracy deviation = 1.957664252% (within 5%).
- Constancy deviations were within tolerance.
- Linearity deviations remained low across the measured range.
- Geometry variation was minimal for the recorded dilutions.

4.6 Precautions

- Wear gloves while handling radiopharmaceuticals and the dose calibrator.
- Carefully place source vial/syringe into the calibrator.
- Use L-bench while filling liquid radioisotope/radiopharmaceutical into syringe or vial.
- Wash hands with soap and water after the experiment.
- Always hold source vials using forceps/holding tongs.
- Switch on fume hood while handling evaporative pharmaceuticals.

5 Conclusion

The quality assurance tests performed on the dose calibrator indicate acceptable instrument performance for routine nuclear medicine activity measurement. Accuracy, constancy, linearity, and geometry-related checks were found to be within acceptable limits based on observed data.

Hence, regular QA of the dose calibrator is essential to ensure reliable radiopharmaceutical activity measurement and safe patient administration.

References

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